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TE-CSA: trajectory entropy-aware cross-step aggregation for improving embedding capacity of distribution-preserving linguistic steganography

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Abstract

Existing diffusion-based linguistic steganography methods largely operate in a timestep-local manner, where embedding decisions are made independently within isolated denoising steps. Such a design fundamentally underutilizes the trajectory-level uncertainty naturally distributed across the diffusion process, causing a substantial gap between achievable embedding performance and the embedding opportunities available throughout the diffusion trajectory. In this paper, we propose TE-CSA, a Trajectory Entropy-aware Cross-Step Aggregation framework for high-capacity distribution-preserving linguistic steganography based on diffusion language models. First, departing from conventional timestep-local embedding paradigms, TE-CSA introduces a trajectory-level aggregation mechanism that jointly exploits embedding opportunities scattered across multiple denoising timesteps. By transforming isolated local candidates into globally coordinated embedding resources, the proposed framework significantly enlarges the effective embedding space and unlocks latent capacity hidden within diffusion trajectories. Second, from an information-theoretic perspective, not all candidate positions contribute equally to embedding. Low-entropy positions often behave as bottlenecks that consume embedding opportunities while providing little effective information gain. To alleviate this entropy bottleneck, TE-CSA incorporates an entropy-aware filtering strategy that selectively retains informative candidate positions and allocates embedding operations to regions with higher entropy concentration, thereby improving effective embedding utilization and enhancing the capacity–quality trade-off. Finally, a distribution-preserving probabilistic sampling mechanism is employed to keep embedding statistically consistent with the original diffusion generation process. Under the assumption that the underlying probabilistic embedding procedure preserves the target token distribution, cross-step aggregation and entropy-aware filtering do not alter the sampling probabilities themselves, thereby retaining the distribution-preserving security property while maintaining natural text quality. We further derive a group-constrained trajectory-level capacity bound for the adopted three-position redundant embedding mechanism and show that capacity-sorted grouping attains this bound under complete trajectory availability. Experimental results demonstrate that TE-CSA achieves up to 18.60× higher embedding capacity and 20.40× higher embedding rate (from 0.09 to 1.84 bits/token) than the architecture-matched timestep-local diffusion baseline, while maintaining limited runtime overhead and generation-quality degradation.

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Keywords Linguistic steganography, Distribution-preserving steganography, Embedding capacity, Entropy utilization, Diffusion models

Introduction

Linguistic steganography conceals secret information in natural language text, enabling covert communication over public channels Provos and Honeyman (2003); Hopper et al. (2002); Fridrich (2009). In a typical communication process, the sender embeds a secret message into seemingly normal text and transmits the generated stegotext through a public channel. The receiver then recovers the hidden message from the received text. Unlike traditional modification-based steganography, generative linguistic steganography embeds information directly during text generation while preserving stegotext distributions that remain statistically close to those of normal language model outputs Yang et al. (2018); Ziegler et al. (2019); Kaptchuk et al. (2021); Cao et al. (2022). This distribution-preserving paradigm has become a fundamental design principle in provably secure linguistic steganography (PSLS). Recent studies also investigate steganalysis robustness using deep learning-based detectors Yang et al. (2019); Peng et al. (2021); Cao et al. (2026).

Recent progress in autoregressive language models (ARMs) and large language models (LLMs) has significantly advanced generative linguistic steganography Vaswani et al. (2017); Radford et al. (2019); Brown et al. (2020); Touvron et al. (2023). Representative autoregressive methods include RNN-Stega Yang et al. (2018), Meteor Kaptchuk et al. (2021), Discop Ding et al. (2023), FreStega Pang (2024), Zero-shot GLS Lin et al. (2024), SparSamp Wang et al. (2025), and CascSamp Lv et al. (2025), which achieve secure or distribution-preserving embedding through probabilistic generation and controlled sampling strategies. Recent studies have further explored adaptive entropy coding and asymmetric numerical systems to improve the efficiency and flexibility of generative linguistic steganography Liu et al. (2025). Recent studies have also extended generative steganography beyond conventional linguistic settings. Zhou et al. (2022) proposed a reversible secret-to-image transformation framework for generative steganography, while subsequent work explored generative steganography through automatically generated semantic object contours Zhou et al. (2023). Recent works published in *Cybersecurity* have further investigated coverless and multi-image steganography Zhang et al. (2025); Ren and Wu (2024). In the linguistic domain, Cao et al. (2022) investigated generative steganography based on long readable text generation. Together with recent neural linguistic steganography methods, these studies demonstrate the increasing diversity of generative information-hiding paradigms. More recently, linguistic steganography has also been extended to large language models, including ChatGPT-based generation frameworks Steinebach (2024).

Beyond autoregressive generation, diffusion language models have introduced a fundamentally different paradigm for text generation. Unlike autoregressive models that generate text through left-to-right token prediction, diffusion language models progressively denoise noisy token states over multiple inference steps. Prior diffusion language modeling studies further explore semantic consistency, controllability, and efficient denoising strategies Lin et al. (2023); Li et al. (2023); Lou et al. (2023). More importantly, the iterative denoising process exposes a sequence of intermediate token distributions rather than a single decoding distribution, creating a trajectory-level probabilistic space in which embedding opportunities naturally emerge across denoising timesteps.

The rapid development of diffusion generation has also motivated increasing interest in diffusion-based steganography Yu et al. (2023); Kim et al. (2025). In the linguistic domain, GTSD Wu et al. (2024) introduced a diffusion-based generative text steganography framework that embeds secret information through prompt mapping and batch mapping. Rather than modifying token-level sampling distributions, GTSD maps secret bits to conditional prompts and selects the final stegotext from batches of diffusion-generated candidate sentences. More recently, STEAD Qi et al. (2026) introduced token-level provably secure linguistic steganography into diffusion language models through step-wise embedding over intermediate denoising distributions.

Recent diffusion-based linguistic steganography studies mainly focus on prompt-level mapping, timestep-local probabilistic sampling, distribution preservation, or robustness enhancement Wu et al. (2024); Qi et al. (2026). Despite these advances, existing diffusion-based linguistic steganography methods still suffer from several limitations: (1) existing token-level diffusion steganography methods such as STEAD perform embedding independently within isolated denoising timesteps, thereby restricting embedding decisions to timestep-local candidate positions. (2) Many candidate positions exhibit highly concentrated

token distributions with extremely low entropy. Directly using such low-capacity positions may reduce effective embedding utilization and introduce unstable distribution perturbations during controlled sampling. (3) Existing methods mainly focus on local sampling correctness and distribution preservation, while the trajectory-level embedding structure naturally contained in diffusion generation remains insufficiently explored.

To address these limitations, we propose TE-CSA, a Trajectory Entropy-aware Cross-Step Aggregation framework for improving embedding capacity of distribution-preserving linguistic steganography. Instead of consuming candidate positions independently within isolated timesteps, TE-CSA dynamically accumulates candidate positions throughout diffusion trajectories and constructs embedding groups across multiple denoising stages. An entropy-aware filtering mechanism selectively removes low-capacity candidate positions before embedding. This improves effective embedding utilization while reducing unnecessary generation-quality degradation.

The main contributions of this paper are summarized as follows:

- We identify inefficient utilization of fragmented candidate positions across denoising timesteps as an important capacity bottleneck in timestep-local diffusion steganography.
- We introduce a cross-step candidate aggregation mechanism that differs from existing timestep-local diffusion steganography by preserving and jointly utilizing candidate positions discovered across multiple denoising timesteps. This mechanism converts temporally fragmented embedding opportunities into globally coordinated embedding groups.
- We further propose an entropy-aware filtering mechanism that distinguishes candidate positions according to their effective embedding potential. Unlike methods that treat all available candidates uniformly, TE-CSA removes low-capacity positions with highly concentrated token distributions before group construction, improving effective embedding utilization and the capacity–quality trade-off.
- We derive a group-constrained trajectory-level embedding capacity bound that explicitly accounts for the three-position redundant embedding mechanism. We further show that capacity-sorted grouping attains this bound when the complete retained candidate set is available, providing a theoretical characterization of the maximum capacity achievable under the adopted grouping constraint.

Background

Diffusion language models

Diffusion language models generate text through an iterative denoising process, where a sequence of latent states $\{x_t\}_{t=0}^{T}$ is progressively refined from noisy representations to coherent natural language outputs. At each denoising step t, the model produces a token-level probability distribution $p_t(v \mid x_t^{(i)})$ for each position i, forming a time-varying probabilistic space over the vocabulary Ho et al. (2020); Austin et al. (2021); Gong et al. (2022).

Recent advances in diffusion language modeling have further improved generation quality and controllability through a variety of modeling paradigms, including self-refined generation, simplified diffusion training objectives, and masked or BERT-style diffusion decoding mechanisms Strudel et al. (2022); Sahoo et al. (2024); He et al. (2023). These studies demonstrate that diffusion language models effectively capture complex semantic distributions while exposing rich intermediate denoising states throughout the generation trajectory.

From the perspective of linguistic steganography, these intermediate distributions define a dynamic sampling space in which embedding operations can be performed. Instead of treating generation as a timestep-local decoding process, diffusion models expose a sequence of evolving probability distributions across timesteps, each corresponding to a distinct embedding opportunity with different entropy characteristics.

As illustrated in Fig. 1, each denoising step induces a token distribution over the vocabulary, and sampling is typically performed via stochastic selection over $p_t(v \mid x_t^{(i)})$. This naturally forms a probability partition structure, which enables information embedding by operating within constrained sampling intervals rather than modifying logits or deterministic outputs.

Existing diffusion-based linguistic steganography methods exploit diffusion generation in different ways. GTSD Wu et al. (2024) performs embedding at the prompt and generated-sentence batch levels, whereas STEAD Qi et al. (2026) operates directly on token-level distributions within individual denoising timesteps. However, neither paradigm explicitly aggregates token-level embedding opportunities distributed across multiple denoising steps.

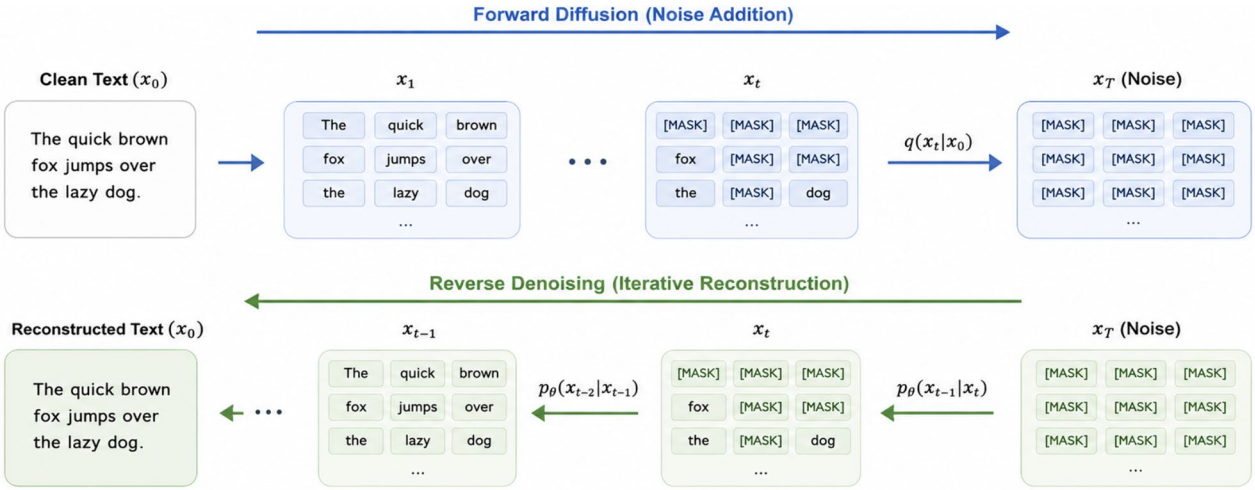


Fig. 1 Diffusion language model viewed from a probabilistic embedding perspective. Each denoising step produces a token distribution $p_t(v \mid x_t^{(i)})$, forming a sequence of evolving sampling spaces across the diffusion trajectory. These intermediate distributions serve as embedding opportunities for diffusion-based linguistic steganography

Information-theoretic capacity

In information theory, entropy measures the uncertainty of a probability distribution and determines its information-carrying capability Hopper et al. (2002); Cachin (1998); Kaptchuk et al. (2021). For a discrete probability distribution $P(v)$, Shannon entropy is defined as:

$$H(P) = -\sum_{v} P(v) \log P(v) \tag{1}$$

Higher entropy generally indicates larger embedding flexibility and stronger information-carrying capability. Existing linguistic steganography studies further demonstrate that embedding capacity in probabilistic text generation is closely related to effective entropy utilization and distribution uncertainty Ziegler et al. (2019); Pang (2024); Qi et al. (2026).

However, directly computing Shannon entropy for every token distribution during diffusion generation can be computationally expensive. Following previous studies, we estimate the integer-valued candidate capacity using a min-entropy-inspired approximation:

$$C(t,i) = \left\lfloor -\log_2 \left(\max_{v} p_t(v \mid x_t^{(i)}) \right) \right\rfloor. \tag{2}$$

where C(t, i) denotes the estimated candidate capacity at denoising step t and position i.

Intuitively, if the maximum token probability is extremely large, the corresponding token distribution is highly concentrated and provides limited embedding flexibility. Conversely, more balanced distributions correspond to larger embedding entropy and potentially larger steganographic embedding capacity.

This observation motivates the entropy-aware filtering mechanism proposed in this paper. Specifically, positions with extremely low embedding capacity values are removed before embedding because they contribute little effective information while increasing distribution imbalance during controlled sampling.

Diffusion-based linguistic steganography

The development of diffusion language models has stimulated growing interest in diffusion-based linguistic steganography. Compared with conventional autoregressive generation, diffusion models expose iterative intermediate states and support partially parallel or batch-level generation, providing new opportunities for information hiding. GTSD Wu et al. (2024) is an early representative diffusion-based generative text steganography framework. It adopts AR-Diffusion as the underlying generation model and embeds secret information through two mapping mechanisms. Prompt mapping maps a segment of secret bits to a conditional prompt, while batch mapping selects the final stegotext from a batch of candidate sentences generated under that prompt. Its embedding capacity is therefore mainly controlled by the prompt-table size, model batch size, and generated sentence length. GTSD demonstrates the potential of diffusion models for improving generation speed, robustness, and embedding flexibility.

More recently, STEAD Qi et al. (2026) introduced token-level provably secure linguistic steganography into diffusion language models. Unlike GTSD's prompt- and

sentence-level mapping paradigm, STEAD performs embedding directly over token distributions exposed during iterative denoising. This provides a more direct connection between diffusion probability distributions and distribution-preserving linguistic steganography.

In parallel, high-capacity provably secure linguistic steganography has also been investigated under autoregressive language models. CascSamp Lv et al. (2025) improves embedding capacity through recursive uniform grouping and cascade sampling. By increasing sampling depth and jointly considering token distributions across consecutive autoregressive prediction steps, CascSamp substantially improves entropy utilization while maintaining distributional consistency.

Despite these advances, the above methods optimize embedding capacity through different mechanisms. CascSamp focuses on improving the utilization of autoregressive conditional distributions, while GTSD embeds information at the prompt and sentence-batch levels. STEAD operates on token-level diffusion distributions but performs embedding within individual denoising timesteps. Consequently, the joint utilization of token-level candidate positions distributed across multiple diffusion timesteps remains insufficiently explored. This observation motivates the trajectory-level cross-step aggregation mechanism proposed in TE-CSA.

The key distinction of TE-CSA is therefore not merely the use of a diffusion language model, but the explicit treatment of the complete denoising trajectory as a shared embedding resource. Existing diffusion-based approaches either operate at the prompt or sentence-batch level, as in GTSD, or perform token-level embedding independently within individual denoising timesteps, as in STEAD. TE-CSA instead introduces a trajectory-level coordination mechanism in which candidate positions discovered at different timesteps can participate in the same embedding process. Entropy-aware filtering further differentiates TE-CSA by explicitly modeling the heterogeneous embedding value of these candidates before group construction.

Proposed method

In this section, we introduce the proposed TE-CSA framework for improving embedding capacity of linguistic steganography in diffusion language models. Unlike existing timestep-local embedding methods, TE-CSA jointly utilizes candidate embedding opportunities distributed across multiple denoising timesteps. The framework combines cross-step aggregation with entropy-aware filtering to improve effective embedding utilization while preserving text quality and distribution consistency.

We first present the overall framework architecture and then describe the core components of the proposed method, including candidate collection, pending-pool aggregation, entropy-aware filtering, and distribution-preserving embedding.

Motivational study

We begin by identifying a fundamental limitation of existing diffusion-based steganography methods through an empirical motivational study.

Existing token-level diffusion linguistic steganography methods such as STEAD generally perform embedding independently at each denoising timestep. Such timestep-local strategies implicitly assume that sufficient embedding opportunities are available whenever embedding is required. To examine whether this assumption holds in practice, we analyze the distribution of candidate embedding positions across diffusion timesteps under the default setting (Length $= 256$, Temp $= 1.3$, and top-$p = 0.98$). For each denoising step, we count the number of candidate positions satisfying the embedding criterion used in STEAD.

As shown in Table 1, a key observation is that embedding opportunities are highly sparse across diffusion trajectories, with the majority of timesteps failing to provide any valid candidate positions. Only a very small fraction of timesteps provide embeddable candidates, and candidate-rich timesteps occur only occasionally throughout the diffusion trajectory.

These results show that candidate positions are sparsely distributed across denoising timesteps. Since timestep-local embedding cannot jointly utilize candidates discovered at different steps, a considerable fraction of otherwise usable embedding opportunities remains under-exploited. This observation motivates the cross-step aggregation mechanism introduced in the following section.

Overall framework

Diffusion language models expose a sequence of evolving token distributions across denoising timesteps, and the corresponding embedding opportunities are unevenly distributed throughout the trajectory. Motivated by the sparse candidate distribution observed in Sect. Motivational

Table 1 Distribution of candidate positions across diffusion timesteps under the default setting

Candidate count	0	1 & 2	≥3
Percentage (%)	92.97	5.66	1.37

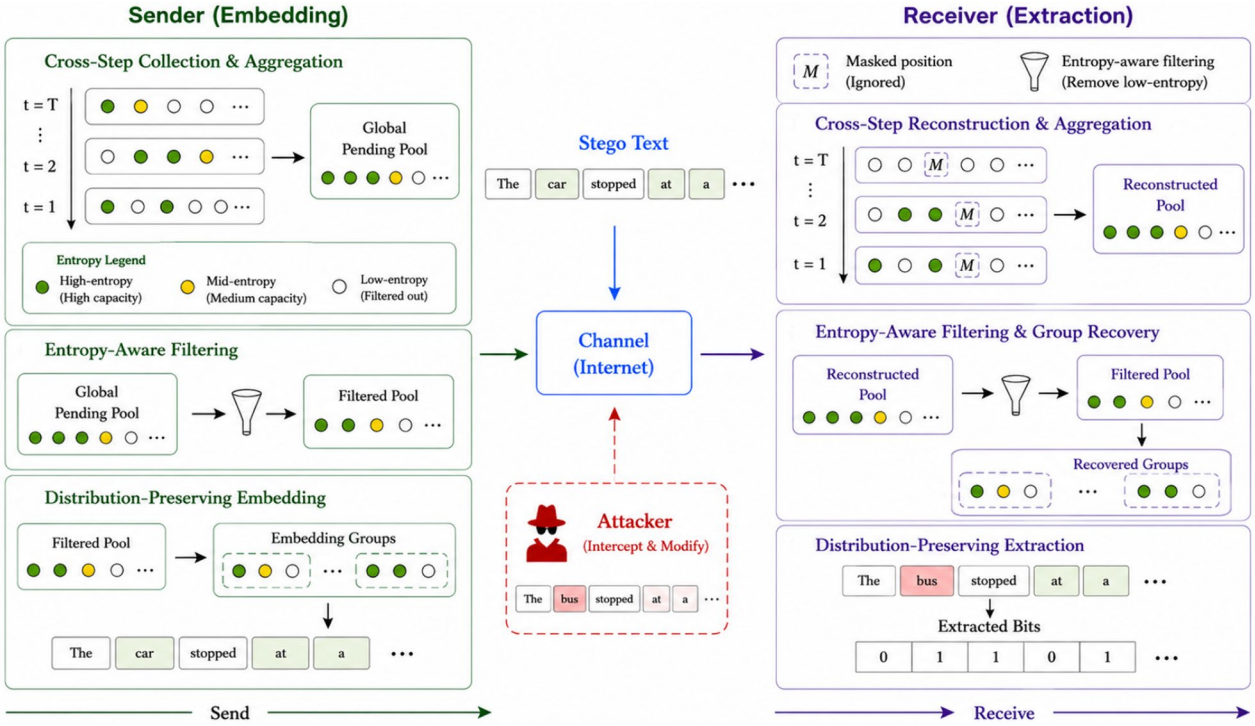


Fig. 2 Overall framework of TE-CSA. The proposed method formulates diffusion-based linguistic steganography as a trajectory-level entropy allocation problem. It integrates cross-step candidate aggregation, entropy-aware filtering, and distribution-preserving embedding into a unified entropy-exploitation pipeline

study, we formulate diffusion-based linguistic steganography as a trajectory-level entropy allocation problem, where candidate-level embedding opportunities from different timesteps are jointly coordinated rather than processed independently.

Based on this perspective, we propose TE-CSA (Trajectory Entropy-aware Cross-Step Aggregation), a unified framework that explicitly models and exploits the trajectory-level entropy structure of diffusion language models for high-capacity distribution-preserving linguistic steganography. The core idea is to transform fragmented candidate-level embedding opportunities across timesteps into a globally coordinated embedding resource through cross-step aggregation and entropy-aware selection mechanisms. Figure 2 presents the overall framework of TE-CSA.

This formulation differs from existing diffusion-based linguistic steganography in two aspects. First, embedding decisions are no longer restricted to candidates available within a single denoising timestep; instead, candidates are accumulated and jointly organized across the trajectory. Second, candidate positions are not treated as equally valuable, but are selectively retained according to their estimated embedding capacity before group construction. These two mechanisms respectively address temporal fragmentation and entropy imbalance, which are not explicitly modeled in existing timestep-local approaches.

The design of TE-CSA is motivated by three fundamental observations on diffusion trajectories:

(1) *Trajectory-level entropy distribution.* Embedding opportunities are not uniformly distributed but instead emerge as sparse high-entropy regions across timesteps.

(2) *Entropy imbalance across candidates.* Different token positions exhibit significantly varying entropy levels, leading to heterogeneous information-carrying capacity.

(3) *Distributional constraint of secure embedding.* Secure probabilistic embedding requires not only sampling within the support of the original diffusion distribution, but also preserving the corresponding token-selection probabilities under the assumed embedding mechanism.

Accordingly, TE-CSA can be viewed as a trajectory-level entropy allocation framework. It first aggregates fragmented entropy signals across timesteps, then

filters low-value entropy regions, and finally converts the remaining high-quality entropy into hidden communication capacity under distribution-preserving constraints.

Cross-step candidate aggregation Candidate positions are continuously extracted from each denoising timestep and accumulated into a global pending pool. This mechanism converts temporally scattered candidate-level embedding opportunities into a unified representation of trajectory-level embedding resources.

Entropy-aware candidate filtering To improve effective embedding utilization, low-information candidates are removed based on their estimated candidate capacities. This step mitigates entropy dilution caused by highly concentrated token distributions and enhances the efficiency of subsequent embedding operations.

Distribution-preserving embedding and extraction Finally, TE-CSA performs probabilistic interval-based embedding over the filtered candidate pool. Under the assumed distribution-preserving embedding mechanism, the cross-step aggregation and filtering operations only determine which candidate positions participate in embedding and do not directly modify their underlying token probabilities. Therefore, these trajectory-level operations do not introduce an additional distributional modification beyond that of the underlying embedding procedure. During extraction, the receiver reconstructs the same candidate sets and probability intervals to recover the embedded message without modifying the received stegotext.

Cross-step aggregation

TE-CSA introduces a cross-step candidate aggregation mechanism that continuously accumulates candidate positions across denoising timesteps and dynamically constructs embedding groups from the aggregated candidates. Unlike timestep-local embedding, which processes candidates independently within each denoising step, cross-step aggregation preserves candidates until sufficient compatible positions are available for group construction.

Because candidate positions are sparsely and unevenly distributed throughout the diffusion trajectory, this accumulate-then-trigger design allows temporally fragmented embedding opportunities to be jointly utilized. The overall aggregation process is illustrated in Fig. 3.

At diffusion timestep t, the language model predicts a token distribution for each masked position:

$$p_t^{(i)}(v) = P(x_i = v \mid x_t) \tag{3}$$

where x_t denotes the current noisy state, i represents the token position, and v denotes a candidate token in the vocabulary. Based on the predicted distribution, TE-CSA evaluates the estimated candidate capacity of each

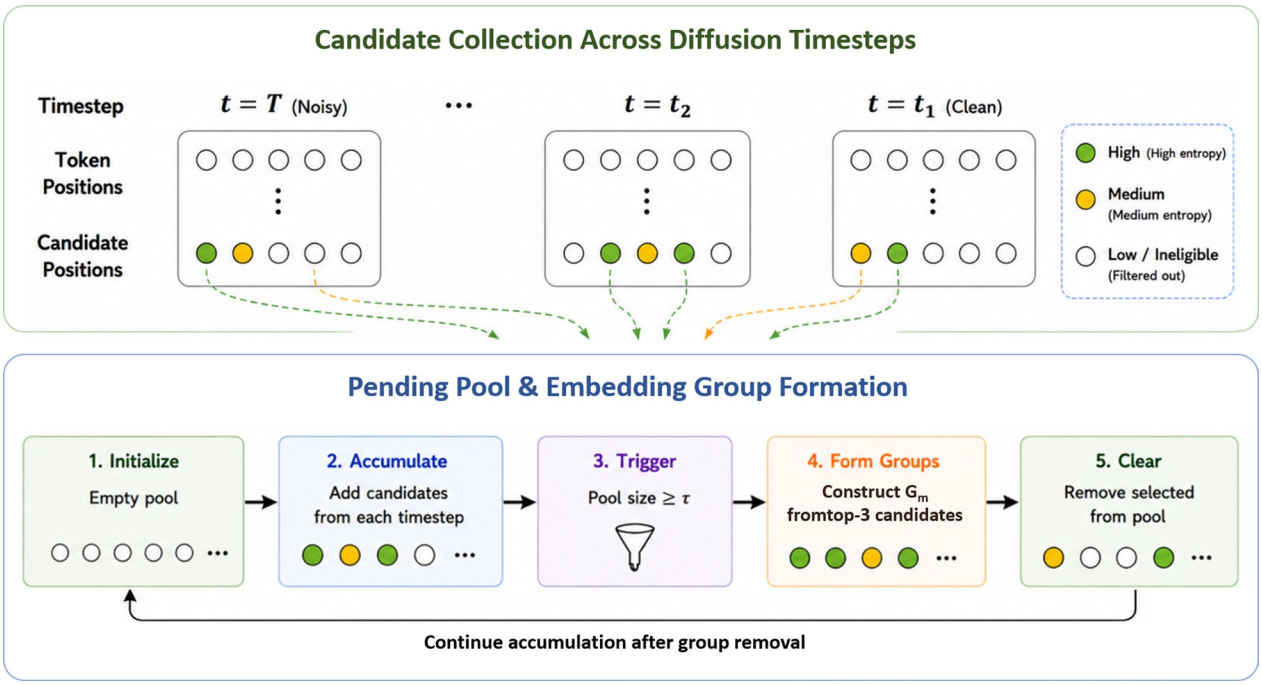


Fig. 3 Cross-step aggregation mechanism in TE-CSA. Candidate positions collected from multiple denoising timesteps are continuously accumulated into a global pending pool. After sufficient high-capacity candidates have been aggregated, the framework dynamically constructs embedding groups for distribution-preserving information embedding

position. Positions satisfying the embedding criteria are added into a candidate set:

$$C_t = \{(t, i) \mid i \in \mathcal{M}_t^{\text{sel}}\} \tag{4}$$

where $\mathcal{M}_t^{\text{sel}}$ denotes the token positions selected for transition at diffusion timestep t. Candidate positions identified at the current timestep are first added to the global pending pool, while their embedding capacities are evaluated subsequently before embedding-group construction.

Unlike conventional timestep-local methods, TE-CSA does not immediately consume candidates from C_t. Instead, all candidate positions are accumulated into a global pending pool:

$$P_t = P_{t+1} \cup C_t \tag{5}$$

where P_t denotes the pending pool after processing timestep t. Intuitively, the pending pool serves as a global candidate reservoir across the diffusion trajectory. Candidate positions discovered at different denoising steps are jointly preserved and later used for embedding group construction. This allows TE-CSA to exploit the full diffusion trajectory rather than isolated timesteps.

As shown in Fig. 3, candidate positions emerge continuously throughout the denoising process. Some timesteps may contain only a few valid candidates, while others may contain more high-capacity positions. Under timestep-local embedding, these sparse candidates are independently consumed at each timestep, resulting in fragmented embedding opportunities and low effective utilization. In contrast, TE-CSA aggregates candidates from multiple timesteps into a unified candidate pool. When the pending pool contains at least three candidate positions, i.e., $|P_t| \geq \tau$ with $\tau = 3$, the framework activates embedding-group construction. The value $\tau = 3$ corresponds to the embedding-group size used in our implementation, where three high-capacity candidate positions are jointly selected for each embedding operation. The aggregation threshold τ controls the minimum number of accumulated candidate positions required before an embedding group can be constructed. A smaller τ triggers embedding more frequently but provides less opportunity for cross-step coordination, whereas a larger τ allows more candidates from different timesteps to be jointly considered at the cost of longer accumulation intervals. This mechanism is referred to as the accumulate-then-trigger strategy. Instead of embedding immediately whenever a candidate appears, TE-CSA waits until sufficient high-quality candidates have been accumulated.

Once triggered, the framework selects a subset of positions from the pending pool to form an embedding group $G_m \subseteq P_t$, where G_m denotes the m-th embedding group. After entropy-aware filtering, the remaining candidates are ranked according to their estimated capacities $C(t, i)$ in descending order. TE-CSA selects the three highest-capacity candidates to construct the embedding group G_m. When multiple candidates have identical capacities, their position indices are used to maintain deterministic ordering. The effective group capacity is determined by the minimum capacity among the three selected positions:

$$C_m = \min_{(t,i) \in G_m} C(t, i), \tag{6}$$

Accordingly, the practical embedding capacity of TE-CSA is determined by the group-level bottleneck capacity rather than by the direct sum of the capacities of all candidate positions. This group constraint is explicitly incorporated into the theoretical analysis in Sect. Group-constrained trajectory-level capacity bound. Because the same secret bit segment is redundantly embedded across the three selected positions, the effective capacity of an embedding group is determined by the candidate with the smallest capacity.

Compared with timestep-local embedding, the proposed aggregation strategy provides two major advantages. First, it significantly increases the number of available high-capacity positions. Since candidate positions are collected across multiple denoising steps, the framework can construct embedding groups with much larger effective capacity. Second, cross-step aggregation improves candidate diversity. Different diffusion timesteps exhibit different uncertainty structures and semantic distributions. Aggregating positions across multiple timesteps allows the framework to better utilize heterogeneous embedding opportunities within the diffusion trajectory. More importantly, the proposed aggregation mechanism reveals that embedding in diffusion language models is inherently a multi-step process rather than a purely timestep-local operation. Figure 3 demonstrates that high-capacity candidates are sparsely distributed across the trajectory, and effective embedding emerges only after sufficient cross-step accumulation. This observation suggests that a major source of the capacity limitation of existing timestep-local methods is the inability to jointly utilize candidate information distributed across multiple denoising steps.

Finally, after embedding group construction, the selected candidate positions are removed from the pending pool:

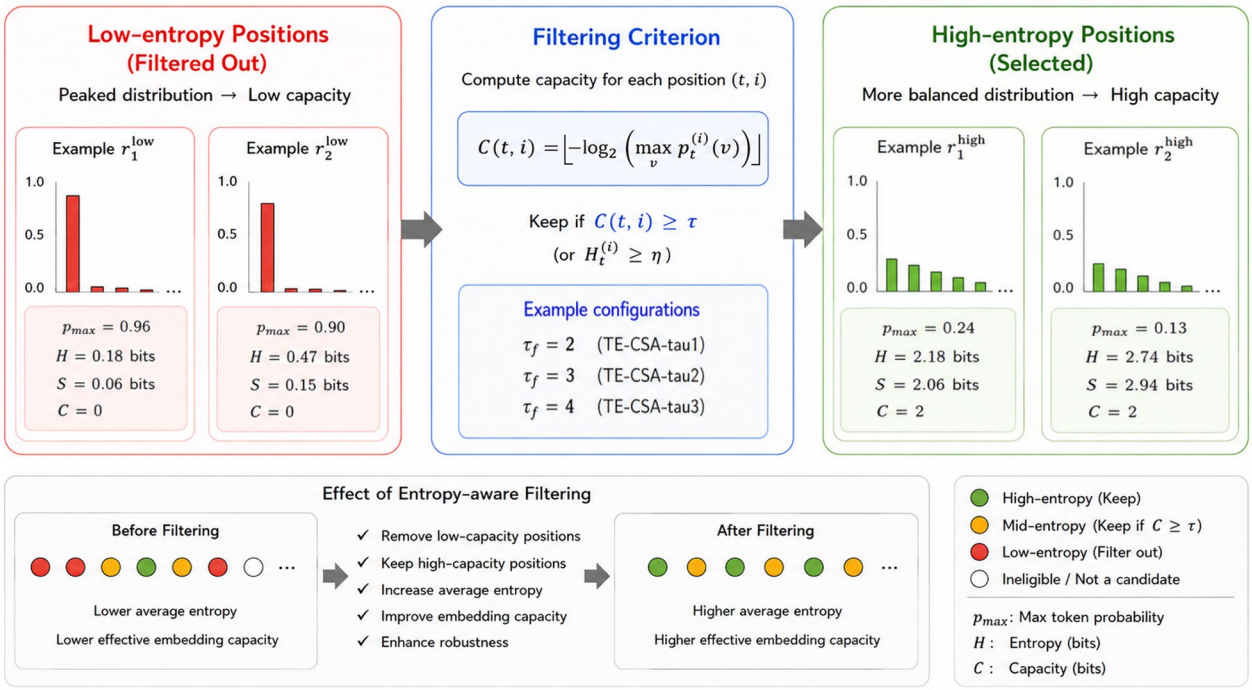


Fig. 4 Entropy-aware candidate filtering mechanism in TE-CSA. Candidate positions collected from multiple denoising timesteps are first evaluated according to their token probability distributions. Low-capacity positions with highly concentrated probabilities are removed from the pending pool, while high-entropy candidates are preserved for subsequent embedding group construction. The filtering process improves effective embedding utilization and generation quality during cross-step embedding

$$P_t \leftarrow P_t \setminus G_m \tag{7}$$

The remaining candidates continue to participate in subsequent aggregation and embedding processes. Through this dynamic accumulation-and-selection mechanism, TE-CSA transforms fragmented timestep-local candidates into globally coordinated embedding groups, thereby considerably improving effective embedding utilization.

Entropy-aware candidate filtering

Although cross-step aggregation increases the number of available candidate positions, their estimated capacities vary substantially because token probability distributions can be highly imbalanced. TE-CSA therefore introduces entropy-aware candidate filtering, which evaluates the estimated candidate capacity of each position and removes low-capacity candidates before embedding-group construction. Positions with highly concentrated token distributions are more likely to provide limited embedding flexibility and are therefore filtered out, allowing the framework to focus on more informative candidates while reducing unnecessary generation perturbation. The overall filtering process is illustrated in Fig. 4.

For a token distribution at timestep t and position i,

$$p_t^{(i)}(v) = P(x_i = v \mid x_t) \tag{8}$$

the uncertainty of the distribution determines its effective embedding flexibility.

According to information theory, the Shannon entropy of a discrete probability distribution is defined as:

$$H_t^{(i)} = -\sum_v p_t^{(i)}(v) \log p_t^{(i)}(v) \tag{9}$$

Higher entropy indicates that the token probabilities are more balanced, implying larger uncertainty and stronger information-carrying capability. Conversely, when the probability mass is concentrated on a small number of dominant tokens, the corresponding entropy becomes extremely low. However, directly computing Shannon entropy for all candidate positions during diffusion generation introduces considerable computational overhead. Therefore, following previous information-theoretic approximations, TE-CSA computes the integer-valued

candidate capacity using the following min-entropy-inspired formulation:

$$C(t,i) = \left\lfloor -\log_2 \left(\max_v p_t^{(i)}(v) \right) \right\rfloor \tag{10}$$

where $C(t, i)$ denotes the estimated candidate capacity of position (t, i). Intuitively, when a token distribution is highly concentrated, the maximum probability approaches 1, i.e., $\max_v p_t^{(i)}(v) \to 1$, which leads to $C(t,i) \to 0$ and indicates extremely limited embedding flexibility.

As shown in Fig. 4, low-capacity positions typically correspond to highly peaked token distributions. Such positions contribute very little effective information because the model is already highly confident about the next token prediction. Embedding information into these positions often requires stronger distribution perturbation, which may lead to larger generation-quality variation while contributing minimal embedding gain.

More importantly, low-capacity positions can directly constrain the effective capacity of an embedding group. As defined in Sect. Cross-step aggregation, the effective group capacity is determined by the minimum candidate capacity within the group. Therefore, if a low-capacity position is included in an otherwise high-capacity group, it becomes the group-level bottleneck and limits the number of secret bits that can be redundantly embedded across the selected positions. Filtering such candidates before group construction can therefore improve the effective utilization of the remaining embedding opportunities.

Based on this principle, TE-CSA removes low-capacity positions before embedding. Specifically, candidate positions are retained only when their estimated capacity exceeds a predefined threshold:

$$P_{\text{filtered}} = \left\{ (t,i) \in P_{\text{cross}} \,\middle|\, C(t,i) \geq \tau_f \right\} \tag{11}$$

where τ_f denotes the numerical filtering threshold. We define three filtering configurations: TE-CSA-τ_1, TE-CSA-τ_2, and TE-CSA-τ_3, corresponding to $\tau_f = 2$, $\tau_f = 3$, and $\tau_f = 4$, respectively.

The filtering process is performed in three steps. First, TE-CSA computes the candidate capacity $C(t, i)$ for each candidate position in the aggregated pool. Second, candidates are ranked or compared according to this value. Third, positions whose capacity is lower than the filtering threshold τ_f are removed before embedding-group construction. Therefore, entropy-aware filtering does not modify the token distribution itself; it only determines whether a candidate position is eligible to participate in subsequent embedding.

In practice, the filtering threshold controls the trade-off between embedding capacity and text quality Ziegler et al. (2019); Kaptchuk et al. (2021). A smaller threshold preserves more candidate positions and may increase total embedding opportunities, but it also introduces more low-entropy positions that contribute limited effective information. Conversely, a larger threshold retains only high-capacity positions, improving embedding efficiency but potentially reducing the total number of available candidates.

To further investigate the influence of low-capacity positions, we evaluate three progressively stronger filtering configurations, τ_1, τ_2, and τ_3, in addition to the no-filter setting. These configurations remove candidate positions with capacities $\{1\}$, $\{1, 2\}$, and $\{1, 2, 3\}$, respectively. Experimental results demonstrate that removing low-capacity positions significantly improves the perplexity-capacity trade-off. In particular, the strongest filtering configuration achieves the best overall performance by eliminating most low-entropy candidates while preserving sufficient embedding diversity.

Another important advantage of entropy-aware filtering is improved distribution stability. Since low-capacity positions usually correspond to highly concentrated token distributions, embedding perturbations at these positions are more likely to reduce generation stability. By removing such positions, TE-CSA reduces unnecessary probability perturbation and helps preserve diffusion sampling consistency.

The proposed filtering mechanism also naturally complements the cross-step aggregation framework introduced in Sect. Overall framework. Cross-step aggregation increases the number of candidate positions across the diffusion trajectory, while entropy-aware filtering further selects the most informative subset among these candidates. Together, these two mechanisms enable TE-CSA to achieve significantly higher embedding efficiency compared with previous diffusion-based steganographic methods. Finally, after filtering, the remaining high-capacity candidates are forwarded to the embedding stage for group construction and distribution-preserving embedding.

Distribution-preserving embedding and extraction

After cross-step aggregation and entropy-aware filtering, TE-CSA obtains a set of high-capacity candidate positions suitable for embedding. The next objective is to embed secret information into these positions while preserving the original diffusion generation distribution as much as possible. Direct token replacement or deterministic modification may introduce obvious statistical artifacts, making generated stegotext vulnerable to steganalysis. Therefore, preserving distribution consistency

during embedding is essential for secure linguistic steganography.

To address this issue, TE-CSA adopts a distribution-preserving probabilistic sampling strategy. Instead of directly modifying generated tokens, the framework embeds secret information through constrained probabilistic sampling over candidate token distributions. Under the assumed distribution-preserving sampling mechanism, this design allows hidden information to be embedded while retaining the target diffusion sampling distribution.

It is important to clarify the security scope considered in this work. TE-CSA does not claim that cross-step aggregation or entropy-aware filtering alone establishes unconditional provable security. Instead, the security property of the framework is inherited from the underlying distribution-preserving probabilistic embedding mechanism. Specifically, cross-step aggregation changes when and how candidate positions are grouped, while entropy-aware filtering determines which candidate positions are retained. Neither operation directly modifies the token probability values produced by the diffusion language model. Therefore, provided that the underlying interval-based embedding procedure samples tokens according to the target diffusion distribution, the additional trajectory-level operations introduced by TE-CSA preserve this distributional property.

Accordingly, we distinguish two levels of security evaluation in this work. The first is conditional distribution-preserving security, which relies on the assumption that the underlying probabilistic embedding procedure faithfully realizes the target token distribution. The second is empirical resistance to practical steganalysis, which is evaluated separately in Section Steganalysis evaluation using neural-network-based detectors. The empirical detection results provide complementary evidence of practical resistance to steganalysis but should not be interpreted as a proof of information-theoretic or universal steganographic security. The overall embedding and extraction process is illustrated in Fig. 5.

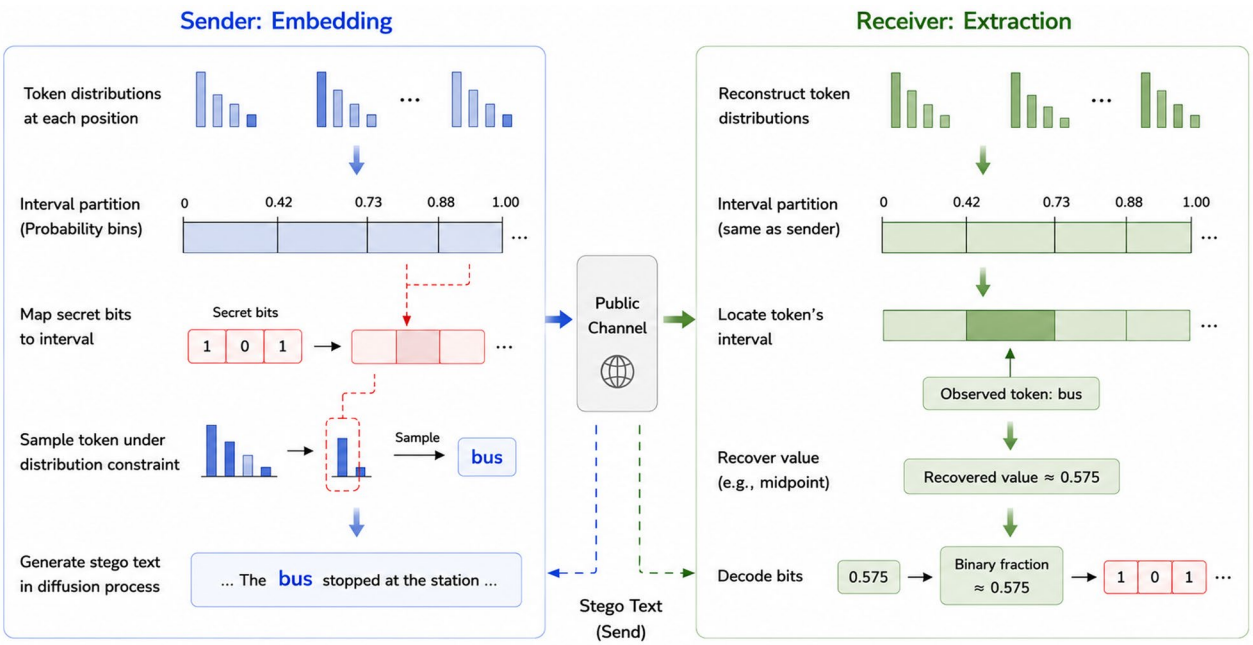


Fig. 5 Distribution-preserving embedding and extraction process in TE-CSA. The sender encodes secret bits into probability intervals derived from diffusion token distributions, while the receiver reconstructs the same intervals to recover the hidden message

Embedding process

Algorithm 1 TE-CSA embedding algorithm

Require: Secret bitstream b; diffusion trajectory $\mathcal{T}$; aggregation threshold $\tau = 3$; filtering threshold τ_f
Ensure: Stegotext x

$P \leftarrow \emptyset$ ▷ Initialize the cross-step pending pool
for each timestep $t \in \mathcal{T}$ **do**
  $C_t \leftarrow \textsc{CollectCandidates}(x_t)$ ▷ Collect candidates at timestep t
  $P \leftarrow P \cup C_t$ ▷ Accumulate candidates across timesteps
  while $|P| \geq \tau$ **do**
    $\{C(t,i)\}_{(t,i)\in P} \leftarrow \textsc{EstimateCapacity}(P)$ ▷ Estimate candidate capacities
    $P^f \leftarrow \textsc{EntropyFilter}(P, \tau_f)$ ▷ Remove low-capacity candidates
    if $|P^f| < \tau$ **then**
      break
    end if
    $P^f \leftarrow \textsc{SortByCapacity}(P^f)$ ▷ Sort by decreasing capacity
    $G_m \leftarrow \textsc{TopCandidates}(P^f, \tau)$ ▷ Select three highest-capacity candidates
    $C_m \leftarrow \min_{(t,i)\in G_m} C(t,i)$ ▷ Determine effective group capacity
    $b_m \leftarrow \textsc{NextBits}(b, C_m)$ ▷ Read the next C_m secret bits
    $z \leftarrow \textsc{BitsToFraction}(b_m)$ ▷ Map bits to a value in $[0,1)$
    $x \leftarrow \textsc{IntervalSample}(G_m, z)$ ▷ Embed b_m over the selected group
    $P \leftarrow P \setminus G_m$ ▷ Remove used candidates
    $b \leftarrow b[C_m + 1 :]$ ▷ Advance the secret bitstream
  end while
end for
return x

Suppose an embedding group G_m contains multiple candidate positions collected across the diffusion trajectory:

$$G_m = \{(t_1, i_1), (t_2, i_2), \ldots, (t_k, i_k)\} \tag{12}$$

For each candidate position, the diffusion language model outputs a token probability distribution:

$$p_t^{(i)}(v) = P(x_i = v \mid x_t) \tag{13}$$

During embedding, the value z is located within one of the token intervals, and the corresponding token is sampled under distribution-preserving constraints. Under the assumed distribution-preserving interval-sampling mechanism, token selection follows the target diffusion distribution. Importantly, embedding is performed only on high-entropy candidate positions aggregated across timesteps, which improves effective embedding utilization and avoids low-capacity regions. The embedding procedure is summarized in Algorithm 1.

Algorithm 2 TE-CSA extraction algorithm

Require: Stegotext x; diffusion trajectory $\mathcal{T}$; aggregation threshold $\tau = 3$; filtering threshold τ_f
Ensure: Recovered bitstream $\hat{b}$
$P \leftarrow \emptyset$ ▷ Initialize the cross-step pending pool
$\hat{b} \leftarrow \emptyset$ ▷ Initialize recovered bitstream
for each timestep $t \in \mathcal{T}$ **do**
  $C_t \leftarrow \textsc{ReconstructCandidates}(x_t)$ ▷ Reconstruct candidates at timestep t
  $P \leftarrow P \cup C_t$ ▷ Accumulate reconstructed candidates
  while $|P| \geq \tau$ **do**
    $\{C(t,i)\}_{(t,i)\in P} \leftarrow \textsc{EstimateCapacity}(P)$ ▷ Recompute candidate capacities
    $P^f \leftarrow \textsc{EntropyFilter}(P, \tau_f)$ ▷ Apply the same filtering rule
    if $|P^f| < \tau$ **then**
      break
    end if
    $P^f \leftarrow \textsc{SortByCapacity}(P^f)$ ▷ Reproduce deterministic ordering
    $G_m \leftarrow \textsc{TopCandidates}(P^f, \tau)$ ▷ Recover the same embedding group
    $C_m \leftarrow \min_{(t,i)\in G_m} C(t,i)$ ▷ Recover effective group capacity
    $I_m \leftarrow \textsc{RecoverIntervals}(G_m, C_m)$ ▷ Reconstruct probability intervals
    $\hat{b}_m \leftarrow \textsc{IntervalDecode}(x, G_m, I_m)$ ▷ Decode the embedded bit segment
    $\hat{b} \leftarrow \hat{b} \| \hat{b}_m$ ▷ Append recovered bits
    $P \leftarrow P \setminus G_m$ ▷ Remove decoded candidates
  end while
end for
return $\hat{b}$

Given a secret bit sequence b, TE-CSA maps it into a continuous value $z \in [0, 1)$ using binary fraction encoding. The token probability space is partitioned into cumulative intervals:

$$I_j = \left[\sum_{k<j} p_k, \sum_{k\leq j} p_k \right) \tag{14}$$

Extraction process

During extraction, the receiver reconstructs the same candidate positions using identical diffusion dynamics and entropy-aware filtering. Since both sender and receiver follow the same aggregation rule, embedding groups can be deterministically recovered.

For each recovered embedding group G_m, the observed token is mapped back to its corresponding probability interval under the reconstructed distribution. The interval index is then decoded into binary representation, yielding the recovered message bits $\hat{b}$. Formally, the decoding process is defined as:

$$\hat{b} = D(G_m) \tag{15}$$

where $D(\cdot)$ denotes the interval decoding function.

Because the receiver reconstructs the same candidate sets, filtering decisions, embedding groups, and probability intervals as the sender, the embedded bitstream can be recovered without modifying the received stegotext. The distributional security property therefore concerns the generation and embedding process rather than the subsequent decoding operation. The extraction procedure is summarized in Algorithm 2.

Theoretical analysis

This section provides theoretical analysis of the proposed TE-CSA framework from two perspectives: (1) group-constrained embedding capacity and (2) computational complexity.

Unlike conventional timestep-local diffusion steganography methods, TE-CSA constructs embedding groups from candidates accumulated across multiple denoising timesteps. Therefore, its practical embedding

capacity depends not only on the candidate-level capacity distributed throughout the trajectory, but also on the three-position grouping constraint introduced by the redundant embedding mechanism.

Group-constrained trajectory-level capacity bound

The original capacity limitation of timestep-local diffusion steganography arises from two distinct factors: the amount of candidate-level embedding potential exposed throughout the denoising trajectory and the structural constraints under which these candidates can be organized into embedding groups. In TE-CSA, the same secret bit segment is redundantly embedded across three candidate positions, and the effective capacity of each embedding group is therefore determined by the minimum capacity among its three members. Consequently, directly summing the capacities of all candidate positions does not represent the practically achievable embedding capacity of TE-CSA. In this section, we explicitly incorporate this group constraint and derive a trajectory-level capacity bound consistent with the actual embedding mechanism.

For each candidate position (t, i), we use the following min-entropy-inspired quantity as a surrogate measure of its candidate-level embedding potential:

$$C(t,i) = \left\lfloor -\log_2 \left(\max_{v} p_t^{(i)}(v) \right) \right\rfloor. \tag{16}$$

where $p_t^{(i)}(v)$ denotes the predicted token probability distribution at timestep t and position i. A larger $C(t, i)$ corresponds to a less concentrated token distribution and therefore indicates greater flexibility for probabilistic embedding.

For any candidate set S, we define its aggregate surrogate capacity budget as

$$B(S) = \sum_{(t,i)\in S} C(t,i). \tag{17}$$

Let

$$P_{\text{traj}} = \bigcup_{t=1}^{T} P_t \tag{18}$$

denote the complete set of candidate positions exposed throughout the diffusion trajectory. The corresponding trajectory-level surrogate budget is

$$B_{\text{traj}} = B(P_{\text{traj}}). \tag{19}$$

Importantly, B_{traj} is neither the Shannon capacity of the complete diffusion process nor the directly achievable embedding capacity of TE-CSA. It only summarizes the candidate-level embedding potential under the surrogate measure in Eq. (16). The actual embedding capacity is further constrained by entropy-aware filtering and by the three-position redundant grouping mechanism.

For a fixed filtering configuration, let

$$P_f = \{(t,i) \in P_{\text{traj}} \mid C(t,i) \geq \tau_f\} \tag{20}$$

denote the retained candidate set, and let $N_f = |P_f|$. Sort the corresponding candidate capacities in non-increasing order:

$$c_1 \geq c_2 \geq \cdots \geq c_{N_f}. \tag{21}$$

Since each TE-CSA embedding group contains three candidate positions and the same secret bit segment is redundantly embedded across all three positions, the effective capacity of a group G_m is

$$C(G_m) = \min_{i\in G_m} c_i. \tag{22}$$

Let

$$M = \left\lfloor \frac{N_f}{3} \right\rfloor \tag{23}$$

denote the maximum number of complete three-position groups that can be constructed from the retained candidate set.

Given the retained candidate capacities in Eq. (21), the maximum total embedding capacity achievable under the three-position redundant grouping constraint is

$$C^*_{\text{GC}} = \sum_{m=1}^{M} c_{3m}. \tag{24}$$

This bound is attained by grouping the sorted candidates consecutively as

$$G^*_m = \{c_{3m-2}, c_{3m-1}, c_{3m}\}, \qquad m = 1, \ldots, M. \tag{25}$$

Proof. Consider any valid partition of the retained candidates into M disjoint three-position groups, and denote the corresponding group capacities, sorted in non-increasing order, by

$$g_1 \geq g_2 \geq \cdots \geq g_M. \tag{26}$$

For the first m groups to each have capacity at least g_m, each of these groups must contain three candidates whose individual capacities are no smaller than g_m. Therefore, at least $3m$ candidates must have capacities no smaller than g_m. Since the candidate capacities are globally ordered as $c_1 \geq c_2 \geq \cdots \geq c_{N_f}$, it follows that

$$g_m \le c_{3m}, \qquad m = 1, \ldots, M. \tag{27}$$

Hence, for any valid grouping,

$$\sum_{m=1}^{M} g_m \le \sum_{m=1}^{M} c_{3m}. \tag{28}$$

The consecutive capacity-sorted grouping in Eq. (25) has group minima exactly equal to $c_3, c_6, \ldots, c_{3M}$ and therefore achieves equality. Thus,

$$C^*_{\mathrm{GC}} = \sum_{m=1}^{M} c_{3m}. \tag{29}$$

□

Equation (24) provides an exact finite-trajectory upper bound under the candidate definition, filtering configuration, and three-position redundant embedding rule adopted by TE-CSA. Compared with the aggregate candidate-level surrogate budget, this bound more accurately characterizes the practically usable embedding capacity because it explicitly accounts for the bottleneck introduced by redundant grouping. In particular, because every valid group contains three candidates whose individual capacities are no smaller than the group minimum,

$$3C^*_{\mathrm{GC}} \le B(P_f), \tag{30}$$

and therefore

$$C^*_{\mathrm{GC}} \le \frac{1}{3} B(P_f) \le \frac{1}{3} B_{\mathrm{traj}}. \tag{31}$$

This relation quantifies the capacity cost introduced by the three-position redundant embedding mechanism and distinguishes the aggregate candidate-level surrogate budget from the practically usable group-level embedding capacity.

The above analysis also provides a direct comparison between timestep-local and cross-step grouping. Let $\mathcal{G}_{\mathrm{local}}$ denote the set of all feasible three-position groupings in which every group is restricted to candidates originating from the same denoising timestep, and let $\mathcal{G}_{\mathrm{cross}}$ denote the corresponding set when candidates from different timesteps may be jointly grouped. Since every timestep-local grouping is also a valid cross-step grouping,

$$\mathcal{G}_{\mathrm{local}} \subseteq \mathcal{G}_{\mathrm{cross}}. \tag{32}$$

Therefore, if C^*_{local} and C^*_{cross} denote the optimal capacities under these two feasible grouping sets, respectively,

$$C^*_{\mathrm{local}} \le C^*_{\mathrm{cross}} = C^*_{\mathrm{GC}}. \tag{33}$$

This inequality formally characterizes the theoretical advantage of relaxing the timestep-local grouping constraint: cross-step aggregation enlarges the feasible grouping space and can therefore utilize combinations of candidate positions that are unavailable to timestep-local embedding.

The actual TE-CSA algorithm operates online: candidates are accumulated in the pending pool and groups are constructed whenever the aggregation condition is satisfied. Let $C_{\text{TE-CSA}}$ denote the realized capacity of this online procedure. Because the online algorithm only observes candidates available up to the current denoising stage, while Eq. (24) assumes that the complete retained candidate set is available before grouping, the realized capacity satisfies

$$C_{\text{TE-CSA}} \le C^*_{\mathrm{GC}}. \tag{34}$$

When the complete retained candidate set is available, capacity-sorted consecutive grouping attains C^*_{GC} exactly. For the online setting, however, equality is not guaranteed because future candidate positions are unavailable at the time an embedding group is triggered.

It is therefore important to distinguish the exact finite-trajectory result established above from the stronger information-theoretic convergence claim made in the original manuscript. The present analysis proves the maximum embedding capacity induced by the adopted candidate-capacity surrogate and three-position grouping constraint under complete trajectory availability. It does not claim convergence to the intrinsic Shannon entropy or information-theoretic channel capacity of the complete diffusion process. A rigorous characterization of such an intrinsic Shannon capacity would require an explicit joint probabilistic model of the complete diffusion trajectory and its inter-timestep dependencies, which is beyond the scope of this work.

Overall, the analysis establishes three conclusions. First, the aggregate candidate-level surrogate budget B_{traj} is not directly achievable because the redundant embedding mechanism introduces a group-level bottleneck. Second, under the adopted three-position grouping rule, the exact complete-trajectory capacity bound is given by C^*_{GC} in Eq. (24), and capacity-sorted grouping attains this bound. Third, cross-step aggregation enlarges the feasible grouping space relative to timestep-local embedding, while the realized online capacity remains upper-bounded by the complete-trajectory group-constrained capacity.

Computational complexity

Let T denote the number of diffusion timesteps, N denote the number of evaluated candidate positions at each

timestep, and K denote the average number of retained candidates after entropy-aware filtering.

Traditional timestep-local diffusion steganography methods independently process candidate positions at each denoising timestep, resulting in computational complexity approximately proportional to $O(TN)$. Compared with previous methods, the proposed TE-CSA framework additionally introduces cross-step candidate accumulation, entropy-aware filtering, and dynamic embedding group construction. However, these operations mainly involve lightweight indexing, candidate management, and local probability comparisons, and therefore do not dominate the overall diffusion inference cost.

During the aggregation stage, candidate positions are continuously inserted into the pending pool:

$$P_t = P_{t+1} \cup C_t \tag{35}$$

where C_t denotes the candidate set collected at timestep t. Since each candidate position is inserted only once and removed after embedding group construction, the overall aggregation complexity remains approximately linear with respect to the number of retained candidates, i.e., approximately proportional to $O(TK)$. Meanwhile, entropy-aware filtering only requires computing a lightweight capacity approximation for each candidate position:

$$C(t,i) = \left\lfloor -\log_2 \left(\max_v p_t^{(i)}(v) \right) \right\rfloor. \tag{36}$$

Unlike full Shannon entropy computation, this formulation only requires identifying the maximum token probability within the local token distribution. Consequently, the filtering process introduces only constant additional overhead per candidate position and remains negligible compared with diffusion model inference itself.

Another important characteristic of TE-CSA is that the framework does not require additional denoising iterations, auxiliary neural networks, or external refinement modules Austin et al. (2021); Hoogeboom et al. (2021). All candidate information is directly reused from the original diffusion generation process. Therefore, the proposed framework effectively improves embedding utilization while preserving the overall inference structure of diffusion language models. In addition, the pending pool size remains dynamically bounded through the accumulate-then-trigger strategy. Candidate positions are continuously consumed after embedding group construction, preventing excessive candidate accumulation during long diffusion trajectories and helping maintain stable runtime behavior throughout generation.

Overall, although TE-CSA introduces additional cross-step management operations compared with conventional timestep-local embedding methods, the overall computational overhead remains relatively small and practical runtime efficiency is largely preserved.

Experiments

To comprehensively evaluate the effectiveness of the proposed TE-CSA framework, we conduct experiments on embedding capacity, embedding efficiency, generation quality, runtime overhead, steganalysis resistance, and cross-step aggregation behavior. Controlled experiments between STEAD and TE-CSA are conducted under unified experimental settings to ensure fair comparison. We additionally report published results of representative high-capacity and diffusion-based linguistic steganography methods for broader methodological comparison; these literature-reported results are explicitly distinguished from our controlled experiments. In the following experiments, Length denotes the generated sequence length, Temp denotes the sampling temperature, and top-p controls the cumulative probability mass retained during nucleus sampling.

Experimental setup

Datasets and Baseline Methods

Following previous diffusion-based linguistic steganography and diffusion text generation studies Gong et al. (2022); Han et al. (2023); Qi et al. (2026), experiments are conducted on open-domain English text generation benchmarks. Prompts are randomly sampled from publicly available datasets covering dialogue generation, open-domain completion, and descriptive generation tasks. The datasets contain diverse sentence structures and semantic distributions, which are suitable for evaluating both generation quality and embedding robustness under diffusion generation settings.

To analyze the influence of generation length on embedding performance, generated text lengths are fixed as Length $\in \{128, 256, 512\}$ throughout experiments. For each configuration, thousands of stegotext samples are generated for statistical evaluation. STEAD and TE-CSA use identical prompts and random seeds in the controlled comparison to ensure fair evaluation.

We consider two types of comparisons: (1) an architecture-matched controlled comparison with STEAD under the same diffusion setting, and (2) a broader literature-based comparison with CascSamp and GTSD to position TE-CSA among representative high-capacity and diffusion-based linguistic steganography methods.

CascSamp Lv et al. (2025) is included as a representative high-capacity provably secure linguistic steganography method based on autoregressive language models. It improves capacity through recursive uniform grouping and cascade sampling. Under its reported LLaMA-7B

setting with top-$p = 0.93$, CascSamp achieves an embedding rate of 3.42 bits/token. Because CascSamp uses an autoregressive backbone and a different evaluation setting, these values are treated as literature-reported references rather than controlled results.

GTSD Wu et al. (2024) is included as a representative diffusion-based generative text steganography method. GTSD embeds information through prompt mapping and batch mapping rather than token-level probabilistic embedding. The original paper reports 1.87−2.22 bits/word under its basic prompt-table configuration and 4.97−5.49 bits/word under its extended configuration across three datasets. Since its capacity depends directly on prompt-table size, model batch size, and sentence length, these results are reported only as literature-based references.

STEAD Qi et al. (2026) is adopted as the primary baseline. STEAD performs embedding independently at each denoising timestep and represents a typical timestep-local diffusion linguistic steganography framework. Since STEAD preserves the original diffusion sampling distribution while achieving provable security, it provides a suitable reference for evaluating the effectiveness of cross-step aggregation and entropy-aware candidate selection. Therefore, STEAD serves as the primary controlled baseline, whereas CascSamp and GTSD provide complementary literature-based references across different high-capacity embedding paradigms. A detailed capacity-oriented comparison is provided in Section Embedding capacity and efficiency.

For fair comparison, STEAD uses the same diffusion backbone, sampling strategy, timestep number, temperature setting, and top-p configuration as TE-CSA. Unless otherwise specified, the default generation setting follows: Length = 256, Temp = 1.3, *and* top-$p = 0.98$.

TE-CSA-no-filter denotes the variant without entropy-aware filtering. All collected candidate positions participate in embedding group construction, allowing us to analyze the influence of low-capacity candidate positions on embedding behavior and overall embedding efficiency. **TE-CSA-** τ_1 , **TE-CSA-** τ_2 , and **TE-CSA-** τ_3 denote progressively stronger entropy-aware filtering configurations. Specifically,TE-CSA-τ_1 removes candidate positions with capacity 1, TE-CSA-τ_2 removes positions with capacities 1 and 2, and TE-CSA-τ_3 removes positions with capacities 1, 2, and 3. Therefore, under TE-CSA-τ_3, only candidate positions with estimated capacity $C(t,i) \geq 4$ are retained for embedding-group construction. Among these variants, TE-CSA-τ_3 achieves the best overall capacity–quality trade-off and is therefore used as the default filtered configuration in the main experiments.

Evaluation metrics

To comprehensively evaluate the proposed framework, we measure embedding performance, generation quality, and computational overhead using the following metrics.

(1) Embedding capacity (C_{Emb} *, bits).* Embedding capacity measures the total amount of hidden information successfully embedded into generated stegotext:

$$C_{\text{Emb}} = \sum_{m=1}^{M} b_m \tag{37}$$

where b_m denotes the number of embedded bits within embedding group G_m. Larger embedding capacity indicates that the framework can utilize more effective embedding opportunities during diffusion generation.

(2) Embedding rate (R_{Emb} *, bits/token).* Embedding rate measures the average number of hidden bits carried by each generated token:

$$R_{\text{Emb}} = \frac{C_{\text{Emb}}}{\text{Length}} \tag{38}$$

Higher embedding rate indicates more efficient utilization of generated tokens for information hiding.

(3) Text quality (γ_{PPL} *, %).* Evaluating text quality in linguistic steganography is challenging because absolute perplexity values are highly dependent on model architecture, sampling strategy, decoding temperature, and generation length. Directly comparing raw perplexity across different settings may therefore introduce significant bias and fail to accurately reflect the actual perturbation introduced by embedding operations.

To provide a more stable and interpretable evaluation, we adopt the relative perplexity variation metric:

$$\gamma_{\text{PPL}} = \frac{\Delta\text{PPL}}{\text{PPL}_{\text{noemb}}} \times 100\% = \frac{\text{PPL}_{\text{stego}} - \text{PPL}_{\text{noemb}}}{\text{PPL}_{\text{noemb}}} \times 100\% \tag{39}$$

where $\text{PPL}_{\text{stego}}$ denotes the perplexity of stego text and $\text{PPL}_{\text{noemb}}$ denotes the perplexity of generation without embedding. Smaller γ_{PPL} values indicate lower distribution perturbation and better preservation of generation quality. Compared with directly reporting absolute perplexity values, γ_{PPL} more effectively reflects the relative influence of embedding operations on the original diffusion generation distribution.

(4) Encoding runtime (T_{Enc} *, seconds).* Encoding runtime measures the average time required to complete stegotext generation and secret information embedding, including candidate collection, cross-step aggregation, entropy-aware filtering, embedding group construction, and probabilistic interval sampling.

(5) Decoding runtime (T_{Dec} *, seconds).* Decoding runtime measures the average time required to reconstruct

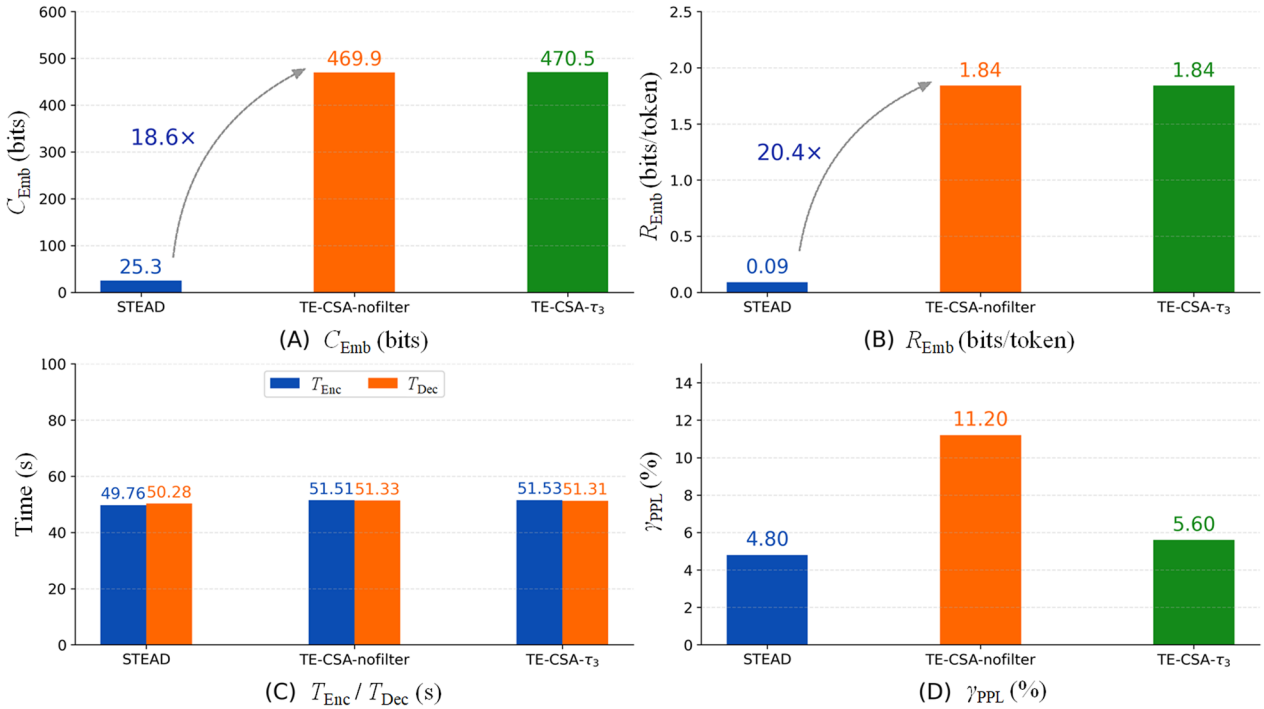


Fig. 6 Overall performance comparison under the default configuration. **A** Embedding capacity. **B** Embedding rate. **C** Encoding and decoding runtime. **D** Text quality measured by γ_{PPL}

Table 2 Controlled performance comparison under different generation lengths

Metric	STEAD Qi et al. (2026)			TE-CSA-no-filter			TE-CSA-τ_3		
Length	128	256	512	128	256	512	128	256	512
C_{Emb} (bits) ↑	12.6	25.3	45.7	190.5	469.9	1010.6	**211.9**	**470.5**	**1022.6**
R_{Emb} (bits/token) ↑	0.09	0.09	0.09	1.49	1.84	1.97	**1.66**	**1.84**	**2.00**
T_{Enc} (s) ↓	32.67	49.76	85.40	32.87	51.51	88.61	33.00	51.53	88.93
T_{Dec} (s) ↓	32.62	50.28	86.75	32.71	51.33	88.43	32.83	51.31	88.60
γ_{PPL} (%) ↓	**4.5**	**4.8**	**4.9**	12.1	11.2	11.8	5.9	5.6	5.6

Bold values denote the best results for (C_{Emb}), (R_{Emb}), and (γ_{PPL}) among the compared methods under the same generation-length setting

embedding groups and recover hidden messages from generated stegotext, including candidate reconstruction, probability interval recovery, embedded token localization, and secret bit extraction.

(6) Steganalysis Detection Error Rate (P_e , %). The detection error rate measures the ability of a steganalyzer to distinguish stegotext from cover text. Following standard linguistic steganalysis evaluation, P_e is defined as

$$P_e = \frac{P_{\text{FA}} + P_{\text{MD}}}{2} \times 100\%, \tag{40}$$

where P_{FA} denotes the probability of falsely classifying cover text as stegotext, and P_{MD} denotes the probability of incorrectly classifying stegotext as cover text. A value approaching 50% indicates that the detector performs close to random guessing and therefore corresponds to stronger practical resistance to steganalysis.

Unless otherwise specified, all reported results are averaged over multiple random seeds under identical experimental settings.

Model parameter settings

Experiments are conducted using a diffusion language model framework following previous work Gong et al. (2022); Qi et al. (2026). Unless otherwise specified, the diffusion process contains $T = 512$ denoising timesteps. The default generation parameters are fixed as Temp $= 1.3$, and top-$p = 0.98$.

Table 3 Embedding-rate and methodological comparison with representative high-capacity linguistic steganography methods

Method	Generation paradigm	Embedding granularity	Capacity mechanism	R_{Emb} (bits/token)
CascSamp Lv et al. (2025)†	Autoregressive	Token / joint-token	Recursive grouping + cascade sampling	3.42
GTSD Wu et al. (2024)†	Diffusion	Prompt / sentence batch	Prompt mapping + batch mapping	1.56–3.94*
STEAD Qi et al. (2026)	Diffusion	Timestep-local token	Step-wise probabilistic embedding	0.09
TE-CSA	Diffusion	Cross-step token	Cross-step aggregation + entropy filtering	1.84

† Results are taken from the original publications and are provided as literature-based references rather than controlled comparisons. * GTSD reports 1.87–2.22 and 4.97–5.49 bits/word under its basic and extended prompt-table configurations, respectively. The displayed bits/token range is approximately normalized from the corresponding three-dataset averages using an empirically estimated conversion factor of 1.34 tokens/word

During diffusion generation, TE-CSA continuously collects candidate positions across denoising timesteps and stores them in a global pending pool:

$$P_t = P_{t+1} \cup C_t \tag{41}$$

When the pending pool size satisfies $|P_t| \geq \tau$, the framework triggers embedding group construction.

To estimate candidate-level embedding potential for filtering, the candidate capacity is approximated as:

$$C(t,i) = \left\lfloor -\log_2 \left(\max_{v} p_t^{(i)}(v) \right) \right\rfloor \tag{42}$$

where low-capacity candidate positions are filtered before embedding execution.

STEAD and TE-CSA use identical prompts, random seeds, diffusion timesteps, and probabilistic sampling strategies to ensure fair comparison. All experiments are conducted on a Linux server equipped with four NVIDIA GeForce RTX 3090 GPUs (24GB memory), Intel Xeon Gold 5218 CPUs, and 128GB RAM. All models are implemented using Python 3.12 with CUDA 12.4. Unless otherwise specified, all experiments are conducted under the same hardware configuration. Results are averaged over 8 independent runs.

In summary, the aggregation threshold is fixed as $\tau = 3$, corresponding to the three-position embedding-group size used throughout our implementation. Entropy-aware filtering is controlled by the configurations τ_1, τ_2, and τ_3, which progressively remove candidate positions with capacities $\{1\}$, $\{1, 2\}$, and $\{1, 2, 3\}$, respectively. Unless otherwise specified, TE-CSA-τ_3 is used as the default filtered configuration, while the filtering strength is varied only in the sensitivity experiment of Section Entropy-aware filtering analysis.

Embedding capacity and efficiency

Figure 6 and Table 2 summarize the controlled comparison between STEAD and TE-CSA variants under the unified diffusion generation setting.

As shown in Fig. 6A, STEAD achieves only 25.30 bits of embedding capacity, while TE-CSA-no-filter and TE-CSA-τ_3 achieve 469.90 bits and 470.50 bits, respectively. The best result corresponds to an 18.60× improvement over STEAD, confirming the capacity limitation imposed by timestep-local candidate utilization.

Figure 6B further shows that TE-CSA markedly improves embedding efficiency. The embedding rate increases from 0.09 bits/token in STEAD to 1.84 bits/token in TE-CSA-τ_3, corresponding to a 20.40× improvement.

The substantial capacity improvement mainly originates from the relaxation of the timestep-local grouping constraint. Under STEAD, candidate positions available at an individual denoising timestep may be insufficient to form an effective embedding group, causing otherwise usable positions to remain under-utilized. TE-CSA preserves these candidates in the pending pool and combines them with compatible candidates appearing at subsequent timesteps. Consequently, the improvement does not arise from creating additional token entropy, but from converting a larger fraction of the embedding opportunities already exposed throughout the diffusion trajectory into effective embedding groups. This mechanism is consistent with the sparse candidate distribution observed in Table 1 and the dynamic aggregation behavior analyzed in Section Cross-step aggregation mechanism.

To provide a broader capacity-oriented comparison, we further compare TE-CSA with STEAD, GTSD, and CascSamp. Table 3 summarizes their generation paradigms, embedding granularities, capacity mechanisms, and embedding rates. Since the methods are evaluated under different architectures and embedding settings, the literature-reported results are provided mainly for methodological and scale reference.

Table 3 shows that TE-CSA achieves a substantial capacity improvement within token-level diffusion steganography, increasing the embedding rate from 0.09 bits/token in the architecture-matched STEAD baseline to 1.84 bits/token. GTSD reports approximately 1.56–3.94 bits/token across its basic and extended prompt-table configurations, with the basic configuration below TE-CSA and the extended configuration achieving a

higher payload. This difference mainly reflects their distinct capacity mechanisms. GTSD embeds information through predefined prompt and sentence-batch mappings, and its payload explicitly depends on the prompt-table capacity, model batch size, and sentence length. Increasing these mapping spaces can therefore directly enlarge its payload, but also requires the sender and receiver to maintain consistent pre-shared prompt-mapping and generation resources; in contrast, TE-CSA improves capacity by utilizing token-level embedding opportunities already exposed throughout the diffusion trajectory without enlarging an external prompt space. CascSamp achieves 3.42 bits/token under an autoregressive generation paradigm, where recursive grouping and cascade sampling can exploit consecutive next-token conditional distributions. Although its absolute rate is higher, TE-CSA reaches a comparable order of magnitude while operating in the more closely constrained token-level diffusion setting. Overall, the comparison indicates that TE-CSA substantially improves capacity within diffusion token-level probabilistic embedding, whereas GTSD and CascSamp obtain high payloads through different embedding spaces and generation paradigms.

Encoding and decoding time

Encoding and decoding efficiency are important factors for practical diffusion-based linguistic steganography. Although TE-CSA introduces cross-step aggregation, entropy-aware filtering, and embedding group construction, these operations remain lightweight compared with diffusion inference itself.

As shown in Fig. 6C, all methods maintain comparable encoding and decoding runtime under the default Length=256 setting. STEAD requires 49.76 seconds for encoding and 50.28 seconds for decoding, while TE-CSA-τ_3 requires 51.53 seconds and 51.31 seconds, respectively. The additional computational overhead remains limited despite the substantial improvement in embedding capacity and embedding rate.

This result mainly arises because TE-CSA does not introduce additional denoising iterations or auxiliary neural network modules. Candidate positions are directly collected from token distributions already produced during the original diffusion process. Therefore, the additional computation mainly comes from lightweight indexing, entropy-aware filtering, pending-pool maintenance, and interval-based probabilistic sampling.

Overall, TE-CSA improves embedding performance primarily through more effective utilization of diffusion trajectory information rather than by increasing generation complexity. Consequently, the proposed framework maintains favorable encoding and decoding efficiency under high-capacity embedding settings.

Text quality

Text quality is evaluated using the relative perplexity variation metric γ_{PPL}, where smaller values indicate lower distribution perturbation introduced during embedding. As shown in Fig. 6D, TE-CSA introduces larger γ_{PPL} values than STEAD because substantially more hidden information is embedded into the generated text. This behavior reflects the inherent trade-off between embedding capacity and generation quality in high-capacity linguistic steganography.

Nevertheless, TE-CSA-τ_3 achieves noticeably smaller γ_{PPL} than TE-CSA-no-filter, reducing the relative perplexity variation from 11.20 to 5.60% while maintaining nearly identical embedding capacity. This result directly validates the effectiveness of entropy-aware filtering. Low-capacity candidate positions usually correspond to highly concentrated token distributions. Embedding information into such positions tends to introduce stronger statistical perturbation because the diffusion model already strongly favors a limited number of candidate tokens. By removing low-entropy candidate positions before embedding, TE-CSA-τ_3 preserves more balanced candidate distributions and reduces unnecessary perturbation to the original diffusion generation process. Therefore, entropy-aware filtering improves the capacity–quality trade-off instead of merely reducing candidate quantity. Quantitatively, filtering reduces γ_{PPL} by 50.0% relative to TE-CSA-no-filter (from 11.20 to 5.60%), while the embedding capacity remains essentially unchanged (469.90 vs. 470.50 bits). This contrast shows that the principal contribution of entropy-aware filtering

Table 4 Steganalysis detection error rate (P_e, %) under different linguistic steganography methods. Values closer to 50% indicate stronger resistance to steganalysis

Detector	CascSamp Lv et al. (2025)†	STEAD Qi et al. (2026)	TE-CSA-no-filter	TE-CSA-τ_3
FCN Yang et al. (2019)	49.73	50.13	49.88	50.10
R-BiLSTM-C Niu et al. (2019)	50.05	51.48	50.21	50.06
BiLSTM-Dense Yang et al. (2020)	49.67	51.77	52.06	51.38

† CascSamp results are taken from the original publication and are provided only as a literature reference because its model and experimental settings differ from those used in our controlled experiments

is not to enlarge the available embedding space, but to remove inefficient low-capacity positions that introduce disproportionate generation perturbation.

Another interesting observation is that the γ_{PPL} value of TE-CSA-τ_3 remains relatively close to STEAD despite achieving nearly 18.6× larger embedding capacity. This suggests that trajectory-level candidate aggregation can effectively increase embedding utilization efficiency without proportionally amplifying distribution distortion.

Overall, these results demonstrate that TE-CSA achieves substantially higher embedding capacity while limiting the additional generation-quality degradation through entropy-aware filtering.

Steganalysis evaluation

To evaluate the practical detectability of the proposed framework, we conduct steganalysis experiments using three representative neural-network-based linguistic steganalyzers: FCN Yang et al. (2019), R-BiLSTM-C Niu et al. (2019), and BiLSTM-Dense Yang et al. (2020). These detectors have been widely used in generative linguistic steganography and are also adopted in the evaluation of CascSamp Lv et al. (2025), providing a useful reference for practical steganalysis resistance.

For the controlled evaluation, cover texts are generated using the same diffusion backbone without secret embedding, while stegotexts are generated using the corresponding steganographic configuration. We evaluate STEAD, TE-CSA-no-filter, and TE-CSA-τ_3 under the default generation setting (Length = 256, Temp = 1.3, and top-p = 0.98). The same prompts and generation settings are used for cover and stego samples to minimize unrelated distribution differences.

Following standard linguistic steganalysis protocols, each detector is trained to distinguish cover texts from stegotexts. We report the detection error rate P_e, where values approaching 50% indicate that the detector performs close to random guessing and therefore suggest stronger practical resistance to steganalysis.

As shown in Table 4, the detection error rates of the evaluated TE-CSA variants remain close to the 50% random-guessing level under all three neural steganalyzers. For TE-CSA-τ_3, the detection error rates obtained by FCN, R-BiLSTM-C, and BiLSTM-Dense are 50.10%, 50.06%, and 51.38%, respectively. These results indicate that the evaluated detectors have limited discrimination capability between TE-CSA stegotext and normal diffusion-generated text under the adopted experimental setting.

TE-CSA-no-filter achieves detection error rates of 49.88%, 50.21%, and 52.06% under the three detectors, respectively. After entropy-aware filtering is applied, the results of TE-CSA-τ_3 remain concentrated around the random-guessing level. Importantly, this practical steganalysis performance is maintained while TE-CSA-τ_3 increases the embedding rate from 0.09 bits/token in STEAD to 1.84 bits/token. These results suggest that the capacity improvement introduced by cross-step

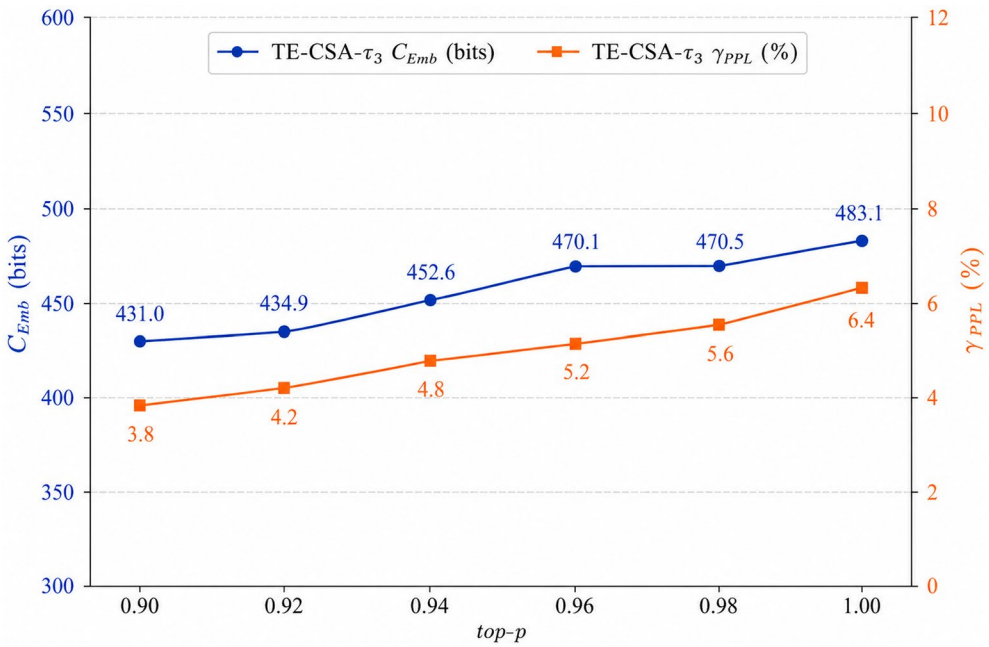


Fig. 7 Influence of different top-p settings on embedding capacity and relative perplexity variation γ_{PPL}

Table 5 Top-*p* sensitivity analysis under different sampling settings

Top-*p*	0.90	0.92	0.94	0.96	0.98	1.00
C_{Emb} (bits) ↑	431.0	434.9	452.6	470.1	470.5	483.1
γ_{PPL} (%) ↓	3.8	4.2	4.8	5.2	5.6	6.4

aggregation does not lead to a substantial increase in practical detectability under the evaluated neural steganalyzers.

For additional reference, Table 4 also reports the published CascSamp results obtained using the same three types of steganalyzers. Under the reported CascSamp setting, the detection error rates are 49.73%, 50.05%, and 49.67% for FCN, R-BiLSTM-C, and BiLSTM-Dense, respectively. Because CascSamp is based on an autoregressive language model and uses different generation and experimental settings, these values are included only as literature-based references and are not treated as controlled numerical comparisons with TE-CSA.

The observed detection resistance of TE-CSA may be related to several characteristics of the proposed framework. First, cross-step aggregation distributes embedding operations across multiple denoising timesteps rather than restricting them to isolated timestep-local candidate groups. Second, entropy-aware filtering avoids low-capacity positions with highly concentrated token distributions. Third, the embedding operation remains constrained by the original diffusion sampling distributions. These properties may help reduce detectable statistical deviations during high-capacity embedding. The empirical steganalysis results therefore complement, rather than replace, the conditional distribution-preserving security argument described in Sect. Distribution-preserving embedding and extraction.

Overall, the steganalysis results provide empirical evidence that TE-CSA remains difficult to distinguish from normal diffusion-generated text under the evaluated neural-network-based steganalyzers while substantially improving embedding capacity and embedding efficiency. Nevertheless, these experiments characterize practical resistance under the selected detectors rather than establishing universal undetectability against all possible steganalyzers. More advanced generalized and LLM-based steganalysis methods Wang et al. (2023, 2025); Yang et al. (2024) remain important directions for future evaluation.

Ablation and mechanism analysis

To further investigate the effectiveness of different components in the proposed TE-CSA framework, we conduct additional ablation and mechanism analysis experiments from three perspectives: (1) sensitivity to the top-*p* sampling parameter, (2) effectiveness of entropy-aware filtering, and (3) dynamic behavior of the cross-step

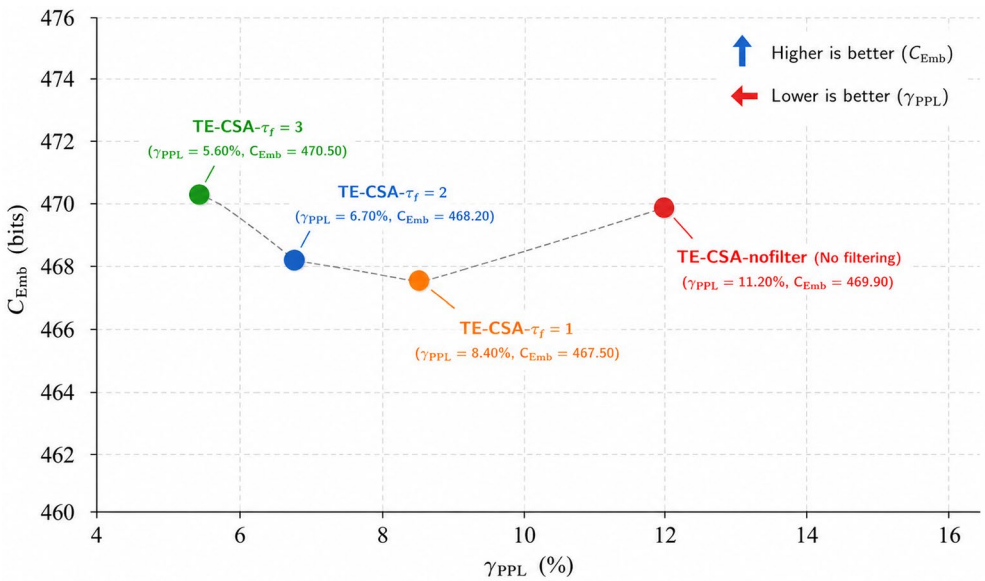


Fig. 8 Influence of entropy-aware filtering configurations on C_{Emb} and relative perplexity variation γ_{PPL}

Table 6 Entropy-aware filtering analysis under different filtering configurations

Configuration	C_{Emb} (bits) ↑	γ_{PPL} (%) ↓
TE-CSA-no-filter	469.90	11.20
TE-CSA-τ_1	467.50	8.40
TE-CSA-τ_2	468.20	6.70
TE-CSA-τ_3	**470.50**	**5.60**

Bold values denote the best result for each corresponding metric among the compared filtering configurations

aggregation mechanism. These experiments aim to analyze how different design choices influence embedding capacity, generation quality, and effective embedding utilization throughout diffusion trajectories.

Top-p sampling sensitivity

In diffusion language models, top-p sampling controls the cumulative probability mass retained during token generation. Larger top-p values preserve more low-probability candidate tokens in the sampling space, thereby increasing generation diversity and potential embedding opportunities. However, excessive sampling flexibility may also introduce larger probabilistic perturbation and reduced generation stability.

Figure 7 and Table 5 summarize the influence of different top-p settings on embedding capacity and text quality under Length = 256 and Temp = 1.3. As shown in Fig. 7, the embedding capacity of TE-CSA-τ_3 gradually increases as top-p becomes larger. When top-$p = 0.90$, the framework already achieves more than 430 bits of embedding capacity. As the sampling space expands, the embedding capacity further increases to 483.1 bits at top-$p = 1.00$.

This behavior indicates that larger top-p values provide richer token diversity and additional embedding opportunities across diffusion trajectories. Since more candidate tokens remain available during probabilistic sampling, TE-CSA can accumulate more valid embedding positions through cross-step aggregation. The proposed framework therefore effectively transforms trajectory-level sampling diversity into usable embedding capacity.

Meanwhile, the relative perplexity variation γ_{PPL} also gradually increases as top-p becomes larger. This trend is expected because larger sampling spaces preserve more low-probability candidate tokens, thereby introducing greater uncertainty during probabilistic embedding. Nevertheless, the increase of γ_{PPL} remains relatively moderate throughout the entire sampling range. Even under top-$p = 1.00$, the relative perplexity variation remains only 6.4%.

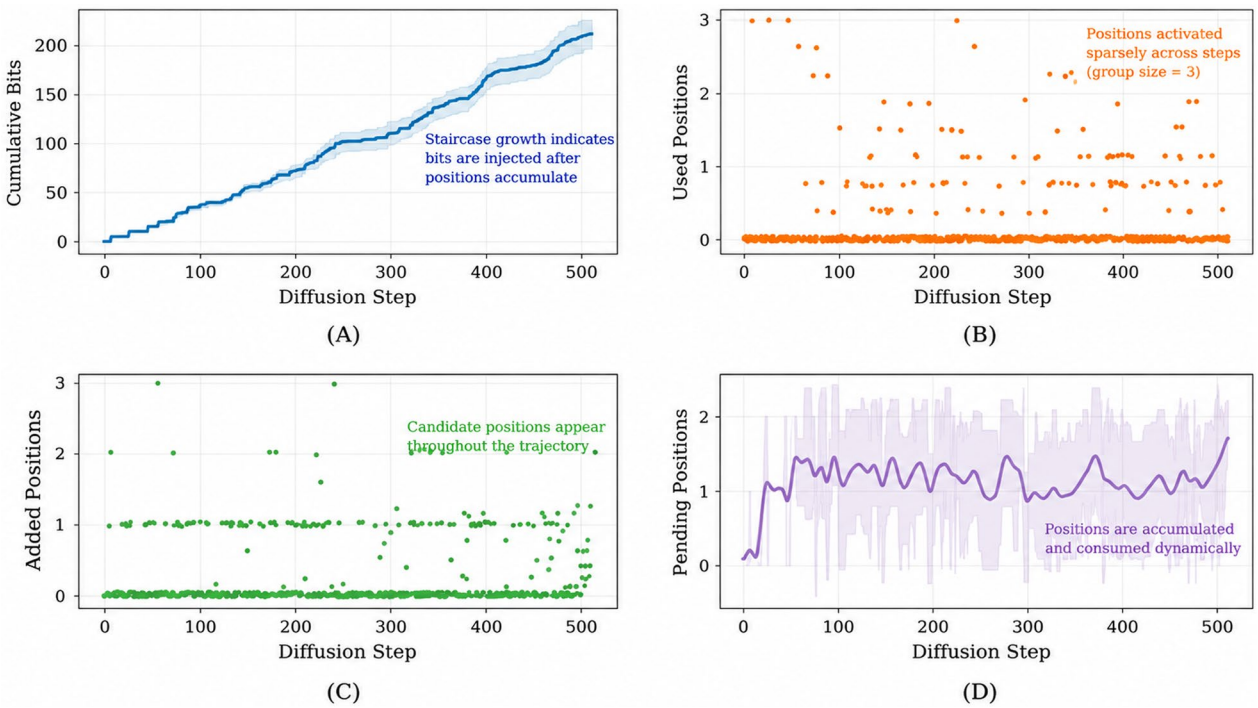


Fig. 9 Cross-step aggregation mechanism analysis. **A** Cumulative embedded bits throughout diffusion timesteps. **B** Activated embedding positions across timesteps. **C** Distribution of candidate positions across denoising trajectories. **D** Dynamic evolution of pending pool size

These results demonstrate that TE-CSA maintains stable embedding behavior under different sampling conditions while effectively utilizing the additional embedding opportunities exposed by larger token sampling spaces. More importantly, the framework preserves a favorable capacity–quality trade-off across a wide range of practical top-p settings.

Entropy-aware filtering analysis

We further analyze the effectiveness of entropy-aware filtering under different filtering configurations. The objective of this experiment is to investigate how entropy-aware candidate selection influences the trade-off between embedding capacity and generation quality.

Figure 8 and Table 6 summarize the experimental results under Length $= 256$, Temp $= 1.3$, and top-$p = 0.98$. As shown in Fig. 8, the embedding capacity remains highly stable across different filtering configurations. Interestingly, TE-CSA-τ_3 slightly outperforms the no-filter setting despite removing multiple low-capacity candidate positions. This observation indicates that low-entropy candidate positions contribute relatively little effective information and may even reduce embedding utilization efficiency.

Meanwhile, the relative perplexity variation γ_{PPL} consistently decreases as the filtering configuration becomes progressively stronger. TE-CSA-no-filter produces the largest perturbation because embedding operations are performed over all candidate positions, including many highly concentrated low-entropy distributions. After progressively filtering these unstable positions, γ_{PPL} decreases from 11.20 to 5.60% while preserving nearly identical embedding capacity.

This behavior demonstrates that entropy-aware filtering effectively improves the capacity–quality trade-off instead of simply reducing embedding quantity. By restricting embedding operations to naturally balanced high-entropy candidate positions, TE-CSA reduces unnecessary probabilistic distortion and preserves stronger consistency with the original diffusion generation distribution.

Overall, these results validate that entropy-aware filtering is not merely a quality-control mechanism, but also an important component for improving effective embedding utilization throughout diffusion trajectories. Together with the TE-CSA-no-filter results, this ablation further separates the contributions of the two major components. Cross-step aggregation alone already increases embedding capacity substantially over the timestep-local baseline, while progressively stronger entropy-aware filtering mainly reduces generation perturbation without sacrificing the capacity gained through aggregation. The final TE-CSA-τ_3 configuration therefore combines the capacity benefit of trajectory-level aggregation with the quality benefit of low-capacity candidate removal.

Cross-step aggregation mechanism

To further understand why TE-CSA achieves substantially larger embedding capacity than timestep-local methods, we analyze the dynamic behavior of the proposed cross-step aggregation mechanism throughout the diffusion process.

Figure 9 visualizes the temporal dynamics of candidate accumulation and embedding execution under Length $= 128$, Temp $= 1.3$, *and* top-$p = 0.98$. As shown in Fig. 9A, cumulative embedded bits exhibit a staircase-like growth pattern rather than smooth linear increase. This phenomenon indicates that embedding execution is triggered only after sufficient candidate positions have been accumulated in the pending pool.

Figure 9B further shows that activated embedding positions are sparsely distributed across diffusion timesteps. Most timesteps contribute only a small number of candidate positions, while a few timesteps provide significantly richer embedding opportunities. This observation directly explains why timestep-local embedding strategies suffer from severe embedding under-utilization.

Meanwhile, Fig. 9C demonstrates that candidate positions continuously emerge throughout the diffusion trajectory rather than concentrating only in the final denoising stages. Therefore, restricting embedding operations to isolated timesteps can leave usable candidate-level embedding opportunities across earlier denoising stages under-utilized.

Finally, Fig. 9D illustrates the dynamic evolution of pending pool size during generation. The pool size gradually increases as new candidate positions are collected and decreases after embedding group construction and candidate consumption. Such behavior demonstrates that TE-CSA dynamically balances candidate accumulation and embedding execution throughout the diffusion process.

Overall, these results provide direct empirical evidence that embedding opportunities in diffusion language models are inherently sparse and cross-step distributed throughout denoising trajectories. Consequently, trajectory-level aggregation plays a critical role in improving effective embedding utilization.

Limitations and scope of application

Although TE-CSA substantially improves embedding capacity by exploiting trajectory-level candidate aggregation, its benefits are accompanied by several practical trade-offs. First, embedding capacity and generation quality are not completely independent. As shown in

Sect. Text quality, increasing the amount of embedded information generally introduces larger distribution perturbations. Entropy-aware filtering alleviates this effect by removing low-capacity positions, but stronger filtering may also reduce the number of available candidates. Therefore, the filtering configuration should be selected according to the desired balance between embedding capacity and generation quality.

Second, TE-CSA introduces additional candidate management operations, including pending-pool maintenance, capacity estimation, filtering, sorting, and embedding-group construction. These operations incur modest computational overhead compared with timestep-local embedding. However, as demonstrated in Sect. Encoding and decoding time, the additional cost remains small relative to the diffusion inference process because TE-CSA does not require extra denoising iterations or auxiliary neural networks.

Third, the effectiveness of TE-CSA depends on the availability of sufficiently diverse candidate positions throughout the diffusion trajectory. The framework is therefore most suitable for diffusion language models that expose multiple intermediate token distributions and exhibit sparse but reusable embedding opportunities across denoising timesteps. In settings with very short trajectories, highly deterministic token distributions, or extremely limited candidate diversity, the advantage of cross-step aggregation may become less pronounced.

Finally, the current implementation uses a fixed aggregation threshold and predefined entropy-filtering configurations. Although these settings provide stable performance under the evaluated generation conditions, adaptive parameter selection according to trajectory statistics may further improve the capacity–quality–efficiency trade-off. Extending TE-CSA to different diffusion architectures, longer text generation, and more dynamic generation settings remains an important direction for future investigation.

Conclusion

This paper proposes TE-CSA, a Trajectory Entropy-aware Cross-Step Aggregation framework for high-capacity distribution-preserving linguistic steganography in diffusion language models. By aggregating candidate positions across denoising timesteps, TE-CSA transforms fragmented timestep-local embedding opportunities into jointly usable trajectory-level resources, while entropy-aware filtering removes low-capacity candidates to improve the capacity–quality trade-off. Under the assumed distribution-preserving probabilistic embedding mechanism, these trajectory-level operations do not directly modify the underlying token probabilities. The theoretical analysis further establishes a group-constrained trajectory-level capacity bound for the adopted three-position redundant embedding mechanism and shows that capacity-sorted grouping attains this bound when the complete retained candidate set is available. Experimental results demonstrate up to an 18.60× improvement in embedding capacity and a 20.40× improvement in embedding rate over the architecture-matched timestep-local diffusion baseline, while maintaining limited runtime overhead and generation-quality degradation. Additional steganalysis and ablation experiments further support the effectiveness of cross-step aggregation and entropy-aware filtering under the evaluated settings.

Future work will focus on improving the adaptability and generalization of TE-CSA. In particular, adaptive aggregation and filtering strategies based on trajectory statistics may further improve the capacity–quality–efficiency trade-off under dynamic generation conditions. Extending the framework to different diffusion language model architectures, longer text generation, and more diverse generation settings will help evaluate its broader applicability. More advanced generalized and LLM-based steganalysis methods may also provide a more comprehensive assessment of practical security. In addition, extending trajectory-level information hiding to multimodal diffusion generation remains an interesting direction for future investigation.

Author Contributions

Runxuan Deng conducted the main research, including methodology development, software implementation, experimental evaluation, and preparation of the manuscript. Xiaowei Yi provided overall supervision, project administration, methodology refinement, funding acquisition, and manuscript review and editing. Hong Zhang, Yun Cao, and Changjun Liu contributed to experimental guidance and manuscript revision. All authors discussed the results, revised the manuscript, and read and approved the final manuscript.

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Data availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Code availability

The source code will be made publicly available in a public repository upon publication.

Declarations

Ethics approval and consent to participate

Not applicable. This study does not involve human participants, human data, human tissue, or animals.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interest.

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